CAN HUMANS COMPETE WITH ARTIFICIAL INTELLIGENCE? THE FUTURE OF WORK AND TASK AUTOMATION

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ABSTRACT

This paper investigates whether artificial intelligence threatens to automate human labor entirely by synthesizing task-based economic theory with recent empirical evidence on AI usage and industrial robotics. Using Acemoglu and Restrepo's displacement-reinstatement framework alongside data from the Anthropic Economic Index and causal estimates from US robot adoption, we analyze five dynamics: the task-job distinction, the race between displacement and reinstatement, the gap between theoretical and observed AI capability, the unprecedented automation of cognitive relative to physical tasks, and emerging policy signals (UBI, AI taxes). We find no evidence supporting near-term mass unemployment, but identify a "so-so automation" risk where task displacement outpaces productivity gains and new task creation. The future of work depends not on technological inevitability but on the direction of innovation and institutional responses to capital-labor substitution.

1 Introduction

Artificial intelligence has advanced to the point where it performs cognitive tasks that once required years of human training writing software, analyzing legal contracts, diagnosing medical images, and even proposing novel mathematical proofs. These capabilities are no longer confined to research laboratories. They are deployed in real workplaces, embedded in enterprise software, and accessible to anyone with an internet connection. The result is a wave of technological change that differs fundamentally from prior industrial revolutions: for the first time in history, automation is attacking cognitive and non-routine white-collar labor before it reaches physical and manual labor. This reversal challenges the historical safety valve that allowed displaced factory workers to migrate into service and knowledge jobs. When the knowledge jobs themselves are the target, the question is no longer which sector will absorb the displaced, but whether the task-based foundation of human employment can regenerate fast enough to keep pace with machine capability.

The central economic insight for understanding this moment comes from the task-based model of the labor market, developed by Autor, Levy, and Murnane and formalized in the displacement-reinstatement framework of Acemoglu and Restrepo. In this view, the economy does not hire "labor" and "capital" as abstract inputs; it assigns a continuum of discrete tasks to the factor human or machine that can perform them at lowest cost. Automation, then, is not the elimination of jobs in one stroke, but the reassignment of tasks within jobs. A programmer does not simply "lose a job" when an LLM writes boilerplate code; the job becomes a bundle of remaining human tasks (system architecture, code review, client communication) and newly augmented tasks (prompt engineering, AI output verification). Whether the job survives depends on whether the remaining tasks are numerous enough, valuable enough, and sufficiently protected from the next wave of automation.

This framework reveals a more precise danger than the popular narrative of "robots taking all jobs." The real risk is what Acemoglu and Johnson term *so-so automation*: technologies that are just productive enough to displace workers and reduce labor costs, but not productive enough to generate the massive prosperity and new labor-intensive tasks that historically compensated for disruption. In the task-based model, automation exerts two opposing forces. The *displacement effect* shifts tasks from labor to capital, reducing the labor share of income and exerting downward pressure

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on wages and employment. The *reinstatement effect* creates new tasks in which labor retains comparative advantage, expanding labor demand and raising the labor share. For much of the twentieth century, reinstatement roughly kept pace with displacement; new occupations (software developer, radiology technician, management analyst) accounted for the majority of employment growth. Since approximately 1990, however, the displacement effect has accelerated while reinstatement has weakened. The labor share of national income has declined, wage growth for middle skill workers has stagnated, and employment polarization has widened signals that the race is tilting in one direction.

This paper examines the future of human labor in an age of generative AI through five interlocking pillars. First, we establish the task versus job distinction as the proper analytical unit, showing why "AI eliminates jobs" is the wrong question and "AI reassigns tasks" is the right one. Second, we model the race between displacement and reinstatement, drawing on Acemoglu and Restrepo's formal framework to identify the conditions under which balanced growth persists and the conditions under which the economy drifts toward a "horse equilibrium" of full automation. Third, we review empirical evidence from the Anthropic Economic Index, which measures the gap between theoretical AI capability and observed professional usage. This data reveals that while large language models could theoretically automate a wide swath of cognitive tasks, actual adoption remains partial, concentrated in programming and administrative work, and largely augmentative rather than fully substitutive. Fourth, we analyze the unprecedented sectoral asymmetry of the current wave. Unlike industrial robots, which displaced physical and routine labor, generative AI targets cognitive, non routine, and often high wage tasks first. This reversal of Moravec's Paradox means that the traditional escape route from factory to office may be closing before new escape routes open. Fifth, we examine policy responses universal basic income, robot taxes, AI dividends, and the EU AI Act not as solutions in themselves, but as implicit signals that industry and government treat tail risk scenarios seriously, even while near-term aggregate unemployment data appears moderate.

Our contribution is synthetic and interpretive. We do not present new primary data, nor do we offer a deterministic forecast. Instead, we construct a coherent analytical framework that connects formal growth theory, causal econometrics from industrial robotics, and real-time AI usage data into a single narrative about the contingency of human employment. We argue that the future of work is not technologically predetermined. It depends on the direction of innovation (whether we build tools that augment labor or machines that replace it), the institutional framework governing capital-labor substitution (whether tax policy subsidizes equipment over employment), and the speed at which new tasks emerge relative to the pace of automation. The evidence does not support predictions of near-term mass technological unemployment; aggregate job destruction on that scale has not materialized, and the reinstatement effect, while weakened, has not disappeared. But the evidence does support a more subtle and perhaps more corrosive concern: a prolonged period in which automation displaces tasks faster than it creates new ones, in which wages stagnate for the cognitive middle class, and in which the gains from AI concentrate among capital owners and a thin slice of highly skilled complementors. The question, then, is not whether humans can compete with artificial intelligence in every domain. The question is whether we will build a society in which humans still have domains worth competing in.

2 Theoretical Framework: The Task-Based Model of Labor

The standard aggregate production function, which treats labor and capital as homogeneous inputs, is ill-suited to analyze automation. It cannot distinguish between a machine that makes a worker more productive and a machine that eliminates the worker's task entirely. To capture this distinction, we adopt the task-based framework developed by Autor, Levy, and Murnane and extended by Acemoglu and Restrepo.

2.1 From Factors to Tasks

In the task-based model, production is not a function of "labor" and "capital" in the abstract. Instead, the economy produces a final good by combining a unit measure of tasks, $y(i)$, indexed along a continuum. Each task can be produced by labor or by capital, depending on technology and relative factor prices. A job, then, is not an input but a bundle of tasks assigned to a particular worker. A software developer, for example, performs a bundle that might include writing code, reviewing colleagues' pull requests, meeting with product managers, and debugging systems. Generative AI may automate the first task without eliminating the job, or it may automate enough tasks that the employer no longer needs the same number of developers.

This re-framing is analytically powerful because it separates two questions that aggregate models conflate: (1) Can a machine perform this specific task? and (2) Is the remaining bundle of tasks still economically viable as a job?

2.2 The Displacement and Reinstatement Effects

Acemoglu and Restrepo introduce two opposing forces that determine the equilibrium path of employment and the labor share.

The *displacement effect* occurs when automation expands the set of tasks that capital can perform. Formally, an increase in the automation threshold I shifts tasks from labor to capital. This reduces the labor share of national income, exerts downward pressure on wages, and may reduce employment if the displaced workers cannot immediately move to other tasks.

The *reinstatement effect* occurs when new technologies create new tasks in which labor has comparative advantage. Formally, an increase in N, the frontier of technologically feasible tasks, expands the set of activities performed by labor. This raises labor demand, increases wages, and expands the labor share.

The net effect of technological change on employment and wages depends on which force dominates. If displacement runs ahead of reinstatement, the labor share falls and workers experience stagnant or declining wages even as average productivity rises. If reinstatement keeps pace, employment and wages can grow alongside automation.

2.3 The Static Model: Intuition

In the static version of Acemoglu and Restrepo's model, the equilibrium allocation of tasks is determined by a threshold I^*. Tasks $i \leq I^*$ are produced by capital; tasks $i > I^*$ are produced by labor. Automation, an exogenous increase in I, always reduces the labor share and employment, and may even reduce wages. Conversely, the creation of new tasks, an increase in N, has the opposite effect.

The critical insight is that automation and new-task creation are not symmetric. Automation directly substitutes capital for labor in existing activities. New-task creation expands the economic pie by introducing activities that did not exist before. This asymmetry means that a wave of "so-so automation," which displaces tasks without creating new ones, can reduce labor demand even while it raises average productivity.

2.4 The Dynamic Model: Balanced Growth

When capital accumulation is endogenous, the model delivers a balanced growth path under specific conditions: exponential productivity growth from new tasks, and equal rates of advancement for automation and new-task technologies. On this path, the rental rate of capital remains constant, and the productivity gains from both types of innovation accrue to labor, the relatively inelastic factor.

The balanced growth path is stable because of a powerful market force: when automation runs ahead of new-task creation, it pushes down the effective cost of producing with labor. This discourages further automation and encourages the creation of new tasks. In other words, the market generates self-correcting tendencies.

However, stability is not guaranteed. If the long-run rental rate of capital is sufficiently low relative to the wage, the incentives to create new tasks disappear, and the economy converges to a "horse equilibrium" of full automation, named after Leontief's analogy that labor may go the way of horses in the twentieth century. The difference between humans and horses, in this framework, is that humans can perform new and more complex tasks; horses could not. Whether this comparative advantage is sufficient to prevent the horse equilibrium depends on the direction of innovation.

2.5 Excessive Automation

In an extension with labor market frictions, Acemoglu and Restrepo identify a new source of inefficiency: *excessive automation*. When the wage rate exceeds the opportunity cost of labor due to minimum wages, unions, or search frictions, firms automate not because it is socially efficient, but because it is privately profitable to avoid labor costs. The social planner, taking into account the lower opportunity cost of labor, would choose less automation. This distortion is exacerbated by tax policies that subsidize capital investment and tax labor through payroll contributions.

3 The Five Pillars of AI Labor Market Impact

We organize our analysis of generative AI's labor market effects around five interlocking pillars. Each pillar addresses a distinct dimension of the problem; together, they form a coherent framework for evaluating whether human labor can remain viable.

3.1 Pillar I: The Task vs. Job Distinction

The first pillar establishes the proper unit of analysis. Public discourse and much media coverage treat "jobs" as the atomic unit of employment, asking whether "AI will replace doctors" or "AI will replace lawyers." The task-based model shows that this is the wrong question.

A job is a bundle of tasks. AI may automate the task of drafting legal contracts, but it does not automate the task of counseling a client through a divorce. It may automate the task of writing boilerplate code, but not the task of translating a business requirement into a system architecture. The economic viability of the job depends on the share of its tasks that remain non-automated, the value of those remaining tasks, and whether new tasks are added to the bundle.

This distinction is not merely semantic. It determines how we measure exposure. Occupational-level studies that ask "Can AI do this job?" overstate displacement risk because they ignore the heterogeneity of tasks within occupations. Task-level analysis reveals that even highly exposed occupations retain significant non-automated components.

3.2 Pillar II: The Race Between Displacement and Reinstatement

The second pillar models the dynamic interaction between task destruction and task creation. Acemoglu and Restrepo's framework predicts that the long-run stability of employment depends on a race: automation destroys tasks, while new technologies create them.

Historically, reinstatement was a powerful force. Between 1980 and 2015, approximately 60 percent of US employment growth came from occupations with new job titles, roles that did not exist in previous decades. These new tasks, software developer, radiology technician, management analyst, social media strategist, absorbed workers displaced from routine manufacturing and clerical jobs.

However, the historical record also reveals a recent imbalance. Since approximately 1990, the displacement effect has accelerated while the reinstatement effect has weakened. The labor share of national income has declined in the United States and other advanced economies. Employment polarization has widened: middle-skill jobs have eroded while high-skill and low-skill employment have grown. These patterns suggest that the self-correcting forces of the model have not operated strongly enough in recent decades to maintain the labor share.

The danger of the current wave is that generative AI accelerates displacement without a corresponding acceleration in reinstatement. Large language models can perform existing cognitive tasks at scale, but they do not automatically invent new labor-intensive tasks at the same rate.

3.3 Pillar III: Empirical Evidence from AI Usage Data

The third pillar grounds the theoretical framework in real-world data from the Anthropic Economic Index. This dataset measures the gap between theoretical AI capability and observed professional usage, a distinction that is often collapsed in popular discourse.

Eloundou et al. measure theoretical exposure (β) by asking whether a large language model could make a task at least twice as fast. Tasks are scored 1 if feasible with an LLM alone, 0.5 if they require additional tools, and 0 if not feasible. The Anthropic Economic Index then tracks actual Claude usage across these tasks, weighting automated uses more heavily than augmentative ones.

The findings are sobering. While 97 percent of observed Claude usage falls into theoretically feasible tasks, the actual coverage of occupational tasks remains a fraction of the theoretical maximum. For example, Claude covers only 33 percent of tasks in the Computer and Math category, despite theoretical models suggesting near-complete feasibility. The most exposed occupations under the observed-exposure metric are computer programmers (74.5 percent), customer service representatives (70.1 percent), and data entry keyers (67.1 percent).

Critically, the data reveal no systematic increase in unemployment for highly exposed workers since late 2022. This suggests that, in the very short run, generative AI is operating primarily as an augmentation tool rather than a displacement shock. However, there is suggestive evidence that hiring of younger workers has slowed in exposed occupations, an early signal that entry-level positions may be contracting as firms substitute AI for junior labor.

3.4 Pillar IV: Sectoral Asymmetry and the Reversal of Automation History

The fourth pillar identifies a structural feature of the current wave that distinguishes it from all prior industrial revolutions. Historically, automation targeted physical, routine, and manual labor: the assembly line, the mechanical loom, the

industrial robot. Workers displaced from factories could migrate into service-sector jobs that required cognitive skills, social intelligence, and non-routine problem-solving.

Generative AI reverses this pattern. It automates cognitive, non-routine, white-collar tasks first, writing, coding, analysis, customer communication, while leaving physical tasks largely untouched. Construction, agriculture, food service, personal care, installation and repair, and protective services remain far less exposed than management, business and finance, computer and math, and legal occupations.

This reversal is an expression of Moravec's Paradox: it is easy to make computers perform tasks that require abstract reasoning, but hard to give them the sensorimotor skills of a one-year-old. AI masters chess and calculus before it masters folding laundry or pruning trees.

The policy implication is profound. The traditional escape route from automation, from physical labor to cognitive labor, may be closing. The remaining less-exposed jobs are disproportionately low-wage physical service jobs. This creates a hollowing-out of the cognitive middle class that is structurally different from the polarization of prior decades.

3.5 Pillar V: Policy Response as Evidence of Tail-Risk Belief

The fifth pillar examines policy proposals not as settled solutions but as revealed beliefs. When serious policymakers, technologists, and economists propose universal basic income, robot taxes, AI dividends, and sovereign wealth funds for AI-generated value, they are implicitly assigning non-trivial probability to scenarios in which market forces alone cannot ensure broad-based prosperity.

The US tax code currently distorts incentives toward automation. Acemoglu and Restrepo note that the tax system aggressively subsidizes the purchase of equipment through accelerated depreciation and investment credits, while taxing labor through payroll taxes. This creates a wedge between the private and social costs of automation, encouraging firms to substitute capital for labor even when the social value of the displaced labor exceeds its private cost to the firm.

Over 60 countries have developed national AI strategies, and the European Union has implemented the AI Act, which classifies AI systems into risk tiers and imposes stricter requirements on high-risk applications. These frameworks are precautionary: they prepare for tail-risk scenarios even while near-term aggregate data appears moderate. The existence of these policy debates is itself evidence that stakeholders do not treat the future of work as a settled question.

4 Causal Evidence from Industrial Robotics

While generative AI is new, industrial robotics provides the best available causal evidence on how automation technologies affect local labor markets. Acemoglu and Restrepo's 2020 study in the *Journal of Political Economy* exploits plausibly exogenous variation in robot adoption across US commuting zones to estimate equilibrium effects.

4.1 Empirical Strategy

The authors construct a Bartik-style measure of exposure to robots. For each commuting zone, they combine baseline industry employment shares with industry-level advances in robotics technology, using European adoption trends as an instrument to isolate the technology-driven component of robot investment. This strategy filters out variation in robot adoption that might be correlated with local labor demand shocks.

The identifying assumption is that commuting zones with higher baseline employment in robot-adopting industries were not differentially exposed to other confounding trends, such as import competition from China, offshoring, or declines in routine tasks. The authors verify this assumption by showing that pre-trends in employment and wages were parallel across high-exposure and low-exposure commuting zones before 1990.

4.2 Magnitude of Effects

The estimates imply robust negative effects of robot exposure on employment and wages. Between 1990 and 2007, the increase in the stock of robots, approximately one additional robot per thousand workers, reduced the employment-to-population ratio in a commuting zone by 0.39 percentage points and average wages by 0.77 percent. These are local effects. Accounting for positive spillovers across commuting zones through trade in goods and shared capital income, the aggregate effects are smaller but still negative: a 0.2 percentage point decline in the employment-to-population ratio and a 0.42 percent decline in wages.

To interpret these magnitudes: one additional robot in a commuting zone reduces employment by approximately 3.3 workers when aggregate spillovers are included, and by about 6 workers when considering local effects alone. The

effects are concentrated in manufacturing, routine manual occupations, blue-collar assembly jobs, and related trades. Negative spillovers extend to nontradable sectors such as construction and retail, as reduced local employment depresses demand for local services.

4.3 Robots Are Not General Capital

A critical finding is that robots have different labor market effects than other forms of capital or technology. The authors find no negative employment or wage effects from overall capital deepening, IT investments, or industry-level productivity growth. This confirms the theoretical distinction between automation technologies, which displace workers from the tasks they were performing, and factor-augmenting technologies, which raise the productivity of labor in its existing tasks.

This distinction is essential for evaluating generative AI. If AI operates primarily as a labor-augmenting tool, making programmers more productive, its effects may resemble capital deepening. If it operates as a task-displacement technology, writing code that previously required a junior developer, its effects may resemble industrial robotics. The evidence from robots suggests that the displacement channel is real, measurable, and economically significant.

5 Synthesis: Can Humans Compete?

Synthesizing the theoretical framework, the five pillars, and the empirical evidence from robotics, we now address the central question: can humans compete with artificial intelligence?

5.1 The Base Case: No Mass Unemployment, But Significant Disruption

The weight of the evidence supports a base-case scenario in which generative AI does not cause near-term mass technological unemployment. Aggregate job destruction on the scale of the Great Depression has not materialized. The Anthropic Economic Index shows that actual AI usage remains partial and augmentative. The reinstatement effect, while weakened, has not disappeared: new occupations continue to emerge, and the task-based model provides a coherent mechanism by which new labor-intensive tasks can stabilize employment in the long run.

However, the absence of mass unemployment is not the absence of harm. The base case is consistent with prolonged wage stagnation for the cognitive middle class, reduced hiring of younger workers in exposed occupations, and a shift in the composition of employment toward low-wage physical services and high-wage AI-complementary roles. The pain is distributional and generational, not aggregate.

5.2 The Pessimistic Case: The So-So Automation Trap

The more troubling scenario is not elimination but erosion. If generative AI continues to displace cognitive tasks without generating sufficient productivity gains or new labor-intensive tasks, the economy may enter a prolonged "Engels' Pause" for the knowledge economy, a period in which output grows but labor income does not. This is the so-so automation trap: technology that is profitable enough to adopt but not transformative enough to compensate the displaced.

In this scenario, the labor share of income continues its decades-long decline. Inequality widens between capital owners and workers, and between AI-complementary high-skill workers and AI-substitutable middle-skill workers. The political and social consequences, declining social mobility, rising populism, collapsing trust in institutions, may exceed the purely economic costs.

5.3 The Tail-Risk Case: AGI and the Breakdown of Models

All of the analysis above assumes narrow AI: systems that perform specific tasks within a bounded domain. If artificial general intelligence were to emerge, systems capable of human-level reasoning across all domains, including the invention of new tasks, the task-based model itself might become obsolete. The comparative advantage of humans in "new and complex tasks" would no longer hold if machines could perform any task, including the creation of tasks.

This scenario is highly uncertain. It lies outside the scope of current empirical evidence and formal economic modeling. But it is the scenario that motivates UBI proposals, AI dividends, and sovereign wealth funds. Policymakers are not preparing for the base case; they are insuring against the tail risk.

5.4 What Survival Means

In the base and pessimistic cases, survival does not mean finding a job that AI cannot do. It means finding work where human judgment, emotional intelligence, physical dexterity, and cross-domain synthesis are complementary to machine capability. The competitive advantage of humans is not raw computation; it is the integration of reasoning with context, ethics, and embodied experience.

The evidence suggests that the most resilient workers will be those who treat AI as a tool rather than a competitor, who use generative models to automate routine components of their jobs while investing in the human-specific tasks that remain. The least resilient will be those who compete with AI on the same task, at the same price, with no complementary advantage.

6 Gaps, Limitations, and Future Research

This paper is synthetic and interpretive; it does not claim to resolve the empirical or theoretical questions it raises. We identify six critical gaps.

6.1 The Developing Country Blind Spot

Nearly all causal evidence on automation comes from the United States and Western Europe. We have almost no rigorous evidence on how generative AI affects labor markets in developing economies, where informality is high, safety nets are weak, and the skill composition of the workforce differs fundamentally. The task-based model may require significant modification to apply to contexts where the formal sector is small and institutional constraints bind tightly.

6.2 The Unpredictability of New Tasks

By definition, we cannot predict what new tasks will emerge. The reinstatement effect depends on innovations that have not yet been imagined. This epistemic limitation is not a flaw in the framework but a boundary condition. Future research should focus on identifying the institutional and technological conditions that facilitate new-task creation, rather than merely forecasting which existing tasks will be automated.

6.3 The AI-Robotics Convergence

The current evidence treats generative AI and industrial robotics as separate phenomena. But the intersection of large language models and physical robots, where AI provides the reasoning and robots provide the embodiment, could eliminate the sectoral asymmetry that currently protects physical labor. When AI can both think and act, Moravec's Paradox may cease to shield construction, agriculture, and personal care. This convergence is barely modeled in the literature.

6.4 Psychological and Social Costs

Economic models measure employment, wages, and the labor share. They do not measure identity, purpose, social cohesion, or the psychological toll of occupational obsolescence. A worker who retains employment but loses the meaning of their work experiences a cost that GDP statistics cannot capture. Future research should integrate sociological and psychological dimensions into the economic analysis of automation.

6.5 The AGI Uncertainty

All existing frameworks assume narrow AI. If AGI arrives, the entire analytical apparatus, tasks, comparative advantage, balanced growth paths, may require fundamental revision. We need theoretical models that can accommodate discontinuities in capability, including scenarios in which machines outperform humans in task creation itself.

6.6 Temporal Mismatch

The Anthropic Economic Index covers approximately two years of data. Labor market adjustments to major technological shocks typically unfold over five to fifteen years. We are observing the early ripples, not the equilibrium. Future research must track occupational transitions, wage dynamics, and task reallocation over longer horizons to distinguish temporary disruption from permanent structural change.

7 Conclusion

The question of whether humans can compete with artificial intelligence is not answered by extrapolating current trends. It is answered by recognizing that the future of work is contingent on choices we have not yet made.

The task-based model provides a rigorous framework for thinking about this contingency. It shows that automation and reinstatement are not natural laws but economic forces shaped by the direction of innovation, the relative price of capital and labor, and the institutional environment. When automation runs ahead of new-task creation, workers suffer. When new tasks emerge faster than old ones are destroyed, workers prosper. Both outcomes are possible; neither is inevitable.

The evidence from industrial robotics proves that automation can cause real, measurable harm to workers in exposed regions and occupations. The evidence from the Anthropic Economic Index shows that generative AI has not yet caused mass unemployment, but has penetrated the cognitive tasks that defined middle-class security. The sectoral asymmetry of this wave, cognitive before physical, breaks the historical pattern that allowed prior generations to escape factory automation by moving into offices. The policy debates over UBI, robot taxes, and AI dividends reveal that serious people treat the tail risk of full automation as non-negligible, even while they disagree on its probability.

We conclude that humans will not be rendered economically obsolete by narrow AI in the near term. But we also conclude that the transition may be prolonged, unequal, and psychologically costly. The danger is not a world without work; it is a world where work pays less, demands more, and concentrates in fewer hands. The question, then, is not whether artificial intelligence will eliminate human labor. It is whether we have the institutional wisdom to ensure that the labor that remains is dignified, well-compensated, and meaningfully human.

The race between man and machine is not a race that technology runs alone. It is a race that we set the rules for.

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A Glossary of Economic Terms

Marginal Productivity The additional output produced by employing one more unit of a factor of production (labor or capital), holding all other inputs constant. In the task-based model, the marginal productivity of labor in a given task determines whether that task is assigned to labor or capital.

Labor Share The fraction of national income that accrues to labor as a factor of production, as opposed to capital. Formally defined as total labor compensation divided by total value added (GDP). A declining labor share indicates that capital is capturing a growing proportion of economic output relative to labor.

CES Production Constant Elasticity of Substitution production function. A functional form used in growth theory that allows the elasticity of substitution between capital and labor to be any constant value. In the task-based model, the aggregate production function is derived from the underlying task structure rather than assumed directly.

Displacement Effect The reduction in labor demand caused by automation technologies that enable capital to perform tasks previously done by workers. In Acemoglu and Restrepo's framework, displacement shifts the task allocation threshold toward labor, reducing employment and the labor share.

Reinstatement Effect The expansion of labor demand caused by the creation of new tasks in which humans retain comparative advantage. Reinstatement counteracts displacement by shifting the task allocation threshold in labor's favor.

So-So Automation A term coined by Acemoglu and Johnson to describe technologies that are sufficiently productive to be adopted and displace workers, but not sufficiently productive to generate large compensatory gains in output or new task creation. Such automation reduces labor demand without proportionally raising aggregate prosperity.

Horse Equilibrium A theoretical outcome in the task-based model where the rental rate of capital falls so low relative to wages that automating all remaining tasks becomes profitable, rendering labor economically obsolete. Named after Leontief's analogy comparing labor to horses in the twentieth century.

Excessive Automation A market failure in which firms automate more than is socially optimal because the private cost of labor (wages plus payroll taxes) exceeds the social opportunity cost of labor (the value of the worker's next-best alternative). Tax distortions and labor market frictions exacerbate this inefficiency.

Task Content of Production The distribution of productive activities across the task continuum, distinguishing between tasks performed by labor and tasks performed by capital. Changes in the task content of production drive shifts in the labor share and employment structure.

Moravec's Paradox The observation, formulated by Hans Moravec, that high-level reasoning requires very little computation, but low-level sensorimotor skills require enormous computational resources. AI systems master chess and calculus before they master folding laundry or walking across a room.

B Data Tables from the Anthropic Economic Index

Table 1: Top 10 Occupations by Observed AI Exposure (Anthropic Economic Index)

Occupation	Observed Exposure (%)	Theoretical Exposure (%)
Computer Programmers	74.5	97.2
Customer Service Representatives	70.1	89.4
Data Entry Keyers	67.1	91.8
Paralegals and Legal Assistants	52.3	85.6
Financial Analysts	48.7	82.1
Market Research Analysts	45.2	79.3
Technical Writers	43.8	88.4
Bookkeeping Clerks	41.5	76.9
Insurance Underwriters	39.2	81.3
Software Developers (Applications)	38.6	94.7

C Timeline of Automation Fears

Table 2: Task Coverage by Major Occupational Category (Observed vs. Theoretical)

Category	**Observed Coverage (%)**	**Theoretical Coverage (%)**
Computer and Mathematical	33.0	91.2
Business and Financial Operations	28.4	76.8
Arts, Design, Entertainment	22.1	68.5
Legal	19.7	72.3
Management	15.2	58.9
Education and Training	8.3	34.6
Healthcare Practitioners	4.1	22.7
Construction and Extraction	1.8	8.4
Food Preparation and Serving	0.9	5.2
Personal Care and Service	0.6	4.1

Table 3: Industrial Robot Effects on US Labor Markets (Acemoglu & Restrepo, 2020)

Effect	**Local (CZ-level)**	**Aggregate (GE)**
Employment-to-population ratio	-0.39 pp	-0.20 pp
Average wages	-0.77%	-0.42%
Workers displaced per robot	≈ 6.0	≈ 3.3

Table 4: Key Moments in the History of Technological Unemployment Anxiety

Year	**Source**	**Claim**
1930	John Maynard Keynes, "Economic Possibilities for Our Grandchildren"	Predicted "technological unemployment" due to our discovery of means of economising the use of labour outrunning the pace at which we can find new uses for labour.
1983	Wassily Leontief, testimony to Congress	Compared labor to horses: horses were made obsolete by the automobile; machines may make human labor obsolete. Introduced the "horse equilibrium" analogy.
2003	Autor, Levy, and Murnane, QJE	Formalized the task-based model showing that computers substitute for routine tasks and complement non-routine tasks. Established the analytical foundation for modern automation theory.
2013	Frey and Osborne, Oxford Martin School	Estimated that 47 percent of US employment is at high risk of computerization. Widely cited but methodologically criticized for overstating occupational-level exposure.
2018	Acemoglu and Restrepo, AER	Provided the first formal model of the race between displacement and reinstatement. Showed that balanced growth is possible but not guaranteed.
2019	Acemoglu and Restrepo, JEP	Documented the recent weakening of the reinstatement effect and the acceleration of displacement. Introduced the "so-so automation" critique.
2020	Acemoglu and Restrepo, JPE	Provided causal evidence that industrial robots reduce local employment and wages. One robot per 1,000 workers reduced employment by 0.2 percentage points.
2023	Acemoglu and Johnson, *Power and Progress*	Argued that technological direction is politically and institutionally contingent, not technologically determined. Emphasized the need for deliberate steering of innovation.
2023	Eloundou et al., OpenAI/UPenn	Estimated that large language models could affect 80 percent of the US workforce, with 19 percent seeing at least half their tasks exposed.
2025	Anthropic Economic Index	Showed that observed AI usage is far below theoretical capability. No systematic unemployment increase yet, but early signals of reduced youth hiring in exposed occupations.